Ventilator Emergencies

AR-09 | Airway & Respiratory | Approved

Approved native content. Beta application; agency release requirements still apply.

Quick flow

01 Ventilator Problem [EMT / AEMT / Paramedic]

Establish baseline: airway type, diagnosis, usual settings, usual SpO_2/$EtCO_2$, alarm meaning, and caregiver rescue plan.

02 Patient Stable? [EMT / AEMT / Paramedic]

If inadequate ventilation/oxygenation or cause unclear: disconnect and BVM immediately.

03 Check Simple Causes [EMT / AEMT / Paramedic]

Oxygen • power/battery • circuit • connection/leak • kink • water • filter • settings/alarm limits.

04 Use DOPE + Auto-PEEP [EMT / AEMT / Paramedic]

Displacement • obstruction • pneumothorax • equipment failure; consider stacked breaths/auto-PEEP.

05 Cause Corrected? [EMT / AEMT / Paramedic]

Yes: resume established settings when safe. No: continue BVM and transport.

06 Suspected TBI? [EMT / AEMT / Paramedic]

SpO_2 ≥94%; urgently correct <90% or inadequate ventilation. Initial/routine assisted $EtCO_2$ 35-45 mmHg. No routine hyperventilation. Clear active/imminent herniation only: brief $EtCO_2$ 30-35 with capnography, avoid <30, and expedite transport. Open TB-05.

07 Transport [EMT / AEMT / Paramedic]

Ventilator • power • battery • oxygen reserve • BVM • suction • spare trach equipment • caregiver when feasible.

Overview

- Treat the patient, not the alarm. Use the caregiver and the patient's functioning equipment whenever possible, but never allow ventilator troubleshooting to delay manual ventilation.
- If oxygenation or ventilation is inadequate, or the cause cannot be immediately corrected, disconnect the ventilator and ventilate with a BVM.

Provider-Level Actions

Use the least invasive effective airway intervention and escalate only when oxygenation or ventilation remains inadequate.

EMT

- Ask the caregiver for the baseline, check power/oxygen/circuit, assess the tracheostomy or ETT, suction when indicated, and immediately use BVM if correction is not rapid.
- Bring the patient's ventilator to the hospital even when malfunctioning.

AEMT

- Perform all EMT care plus advanced suctioning, tracheostomy/BIAD support, capnography, vascular access when indicated, and assistance with ventilator troubleshooting.

Paramedic

- Perform all prior care, maintain functioning home settings when appropriate, and manage airway, pneumothorax, bronchospasm, auto-PEEP, or other physiologic cause.

- Transport on the patient's functioning ventilator with current settings when safe; do not make unneeded setting changes during transport.

Indications

- Ventilator-dependent patient with alarm, equipment failure, hypoxia, abnormal $EtCO_2$, respiratory distress, or suspected airway/circuit problem.

Treatment steps

- Establish the baseline: ventilator-dependent diagnosis, tracheostomy versus ETT, usual settings, usual SpO_2/$EtCO_2$, alarm meaning, and caregiver rescue plan.
- If the patient is stable, use the functioning ventilator and established settings. If oxygenation/ventilation is inadequate or the cause is not immediately clear, disconnect the ventilator and manually ventilate with BVM.
- Check oxygen source/pressure, battery or AC power, circuit connection, leaks, kinks, water, filter, settings, and alarm limits.
- Use DOPE for deterioration: displacement, obstruction, pneumothorax, equipment failure. For asthma/COPD or prolonged exhalation, consider stacked breaths/auto-PEEP; disconnect, manually ventilate slowly, and allow full exhalation before restarting.
- For high-pressure alarm, assess kinking, biting, secretions/plugging, bronchospasm, pneumothorax, and patient-ventilator dyssynchrony. For low-pressure alarm, assess disconnection, circuit leak, cuff leak, loose tubing, or displaced airway.
- Suction when secretions or obstruction are suspected; preoxygenate when possible and never force the catheter.
- During BVM rescue, use age-appropriate rate, visible chest rise, and $EtCO_2$/clinical response; avoid hyperventilation.
- Maintain continuous SpO_2 and waveform $EtCO_2$ when available. Reassess color, chest rise, breath sounds, airway depth, and perfusion after every intervention.
- Transport with the ventilator, power cord, backup battery, adequate oxygen reserve, BVM, suction, and patient-specific spare tracheostomy equipment when available. Bring the knowledgeable caregiver and ventilator settings/information when feasible.
- For suspected traumatic brain injury, continuously monitor oxygenation and ventilation. Aim for SpO_2 at least 94%; urgently correct SpO_2 below 90% or inadequate ventilation. The condition-specific TBI target takes priority over a lower general airway or post-ROSC oxygenation target.
- For suspected traumatic brain injury requiring assisted ventilation, use an initial/routine $EtCO_2$ target of 35-45 mmHg. Avoid routine hyperventilation. Only for clear active or imminent cerebral herniation, use brief controlled ventilation targeting $EtCO_2$ 30-35 mmHg with continuous waveform capnography while expediting definitive care; avoid overshooting below 30 mmHg.
- Prevent hypotension in suspected traumatic brain injury; do not use permissive hypotension. For adults, target SBP at least 110 mmHg, including age 70, with frequent reassessment and after treatment. Pediatric TBI blood-pressure targets require age-specific direction from Medical Control until the separately approved pediatric TBI target table is published; do not substitute adult targets or generic pediatric shock thresholds.

Warnings / precautions

- Never delay manual ventilation for troubleshooting. If the patient improves after disconnection, reassess for ventilator/circuit malfunction, obstruction, or auto-PEEP before reconnecting.
- Do not make unneeded setting changes when the patient is stable on a functioning home ventilator.
- Continuous SpO_2 and waveform $EtCO_2$ are required during assessment and transport once available.

Clinical pearls

- High-pressure and low-pressure alarms suggest different causes; assess the patient and airway before focusing on the machine.
- A ventilator malfunction does not replace the need to assess airway displacement, secretions, pneumothorax, or worsening underlying physiology.

Related pathways / contacts

App links open the deployed version. Confirm its package version matches this PDF before use.

AR-08 Post-Intubation / BIAD Management - Open for recently placed ETT/BIAD confirmation, monitoring, and DOPE response.

AR-10 Tracheostomy Tube Emergencies - Open for tracheostomy obstruction, displacement, or decannulation.

AR-11 Mechanical Ventilation, Adult - Open for adult ventilator setup and management after the emergency is stabilized.

TB-05 Head Trauma - Open the shared TBI oxygenation, ventilation, herniation-rescue, and perfusion targets.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS formulary, linked protocol pathways, and agency release guidance.

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